

2 Units

2.1 Guidance on unit content

Use of i.e./e.g. in unit content

The unit content describes what has to be taught to ensure that learners are able to access the highest marks.

Anything which follows an i.e. details what must be taught as part of that area of content.

Anything which follows an e.g. is illustrative, it should be noted that where e.g. is used, learners must know and be able to apply relevant examples in their work, though these do not need to be the same ones provided in the unit content.

Teachers will need to ensure that any modifications to tasks, from the bank of set assignments for the optional units, do not expect the learner to do more than they have been taught, but they must enable them to access the full range of marks as described in the marking criteria.

For externally assessed units, where the content contains i.e. and e.g. under specific areas of content, the following rules will be adhered to when setting questions:

- direct questions may be asked where the unit content is shown with an i.e.
- where unit content is shown as an e.g., a direct question will not be asked about that example. Any questions relating to the area of content will offer learners the opportunity to provide their own examples, as the unit has not specified which examples they should be familiar with.

2.2 Unit R001: *Understanding computer systems*

Aims

This unit will provide learners with the underpinning knowledge and understanding required to use computer systems effectively. Learners will develop their knowledge and understanding of the systems they use both at home and at school and will explore how these same technologies are used by business organisations.

This unit complements unit R002. In unit R001 learners will study the computer system on which applications software sits and consider the implications of working with data to create content, while in unit R002 they will work with 'office' applications software to edit and format/create content to meet specified business purposes.

From personal computers to smartphones, computing devices are an essential feature of the modern world. Technology may be changing every day, but the knowledge and understanding of how to use computers effectively is the same regardless of the technology being used. Computers are powerful devices for the storage and manipulation of data, but how can they be used effectively and the important data they use be stored securely?

On completion of this unit, learners will have gained the knowledge and understanding to use computers more effectively in a variety of different contexts including home, school and the workplace. Their regard for their own personal data security and for the security of the data of others will be increased and, overall, learners will be more informed users of computers making them more effective participants in business and social life.

Learners studying optional units will be able to apply the knowledge and understanding they have gained in this unit to develop their skills.

Learning Outcome 1: Understand how ICT can be used to meet business needs

Learners must be taught:

- features and purposes of computing devices, i.e.:
 - o desktop and portable devices, i.e. laptops, netbooks, tablets, smartphones
 - o input devices, i.e. mice, keyboard, microphone, sensors, pads, specialist keyboards, touch pad, microphones, remote controls, scanners, digital cameras, webcams, touch screens, readers for barcodes, magnetic stripes and chip and pin, MIDI instruments
 - o output devices, i.e. monitor/screens, printers, speakers, head/earphones, digital projectors, data projectors, plotters, actuators
 - o software, i.e.:
 - operating systems (e.g. Windows, OS X, Android, iOS)
 - utility software (e.g. computer security)
 - applications software, i.e. word processors, desktop publishing software, spreadsheets, database management software, multimedia software, slideshow software, video-editing software, graphics manipulation software, communications software (e.g. social networking software, chat, instant messaging, file transfer and email clients), presentation software, gaming software, web browsers, apps for portable devices
 - o storage and connectivity devices, i.e.:
 - optical disks (e.g. CD and DVD for data storage)
 - magnetic media (e.g. internal and external hard disk drives (HDD), tape)
 - Solid State Drives (SSD)
 - Memory cards, i.e. flash memory devices

- network devices (e.g. routers, modems)
- cloud storage
- o configurations, i.e.:
 - typical office configurations
 - customised systems for specified needs, i.e.:
 - physical impairment, i.e. sight, hearing, movement
 - remote working (e.g. travelling, hotel or home)
- how the following factors can affect the choice of system: cost, availability, user needs, data security
- how peripherals can be connected to a computer device, i.e.:
 - o wired methods (e.g. USB, firewire)
 - o wireless methods (e.g. wifi, Bluetooth, infra-red)
- how to connect a computing device to an existing wireless network, i.e.:
 - o network name, i.e. Service Set Identifier (SSID)
 - o the use of security keys
 - o appropriate firewall settings for public and private networks
- how organisations can monitor employees, i.e. GPS location tracking, monitoring internet use, monitoring communications.

Learning Outcome 2: Know how to work with information and data to meet specified business needs

Learners must be taught:

- data capture methods, i.e.:
 - o online and paper-based forms
 - o automated data capture systems, i.e. control system sensors, barcode readers, Radio Frequency Identification Device (RFID), Near Field Communication (NFC)
- how the following factors can affect the choice of method:
 - o nature of information to be collected (e.g. environmental conditions, location of information)
 - o cost
 - o availability
 - o ease of use
 - o data security
- how to design data capture forms to obtain specified information
- how to code information for use in a spreadsheet or database
- data validation methods
- file formats for storing data, i.e.:
 - o proprietary formats, i.e. .doc, .xls, .ppt, .fla, .wma, .aac
 - o open formats, i.e. .rtf, .pdf, .csv, .exe, .txt, .mp3, .wav
- data storage technologies, i.e.:
 - o local and removable media
 - o remote storage (e.g. offsite location, cloud storage)
- security measures to be used when storing data, i.e.
 - o network/computer security, i.e.:
 - usernames/passwords
 - access rights/permissions
 - o document security, i.e.:
 - passwords
 - other ways to restrict access to or editing of content

- how and why data is encrypted
 - o physical security to prevent loss of data/devices (e.g. locked doors)
- data transferring technologies, i.e.:
 - o wired and wireless methods
 - o mobile data transmission (e.g. 3g, 4g)
 - o remote methods (e.g. email, internet/cloud, peer to peer file sharing)
 - o security methods, i.e. data encryption
 - o how the following factors can affect the choice of method: file size, transfer speed, future-proofing, data security, user needs
- factors affecting data transfer speed (e.g. bandwidth, router technology)
- the factors affecting the appropriate optimisation of electronic files (e.g. download speeds, quality of product)
- how to use back-up and recovery systems, i.e.:
 - o data storage media (e.g. removable devices, remote storage)
 - o back-up frequency
 - o archiving
 - o automated versus manual systems
- how the following factors can affect the choice of method: cost, availability, ease of use, data security.

Learning Outcome 3: Know how ICT can be used to support business working practices

Learners must be taught:

- how businesses can communicate with employees and others working remotely, i.e. voice telephones, SMS, instant messaging, e-mail, chat rooms, forums, bulletin boards, Voice-over-IP (VoIP), video conferencing, webcams, blogs, social networking
 - o appropriate use of remote communication tools, i.e. for email appropriate use of subject, cc/bcc, attachments and email etiquette
 - o the benefits and drawbacks of these methods
- how diary management software can be used to organise work schedules, i.e.:
 - o creating appointments/meetings
 - o inviting participants
 - o creating tasks
 - o creating to-do lists
 - o setting reminders
- how documents can be created and edited collaboratively, i.e.:
 - o documents in shared access locations, i.e.:
 - network shared areas (e.g. read/write access)
 - cloud-based services (e.g. providing open or restricted access to services enabling the creating/editing of documents online)
 - o inserting comments into an existing draft
 - o editing drafts, tracking changes made
 - o reviewing facilities: accepting or rejecting changes made.

Learners must be taught:

- how legislation affects business computer users, i.e.:
 - o health and safety
 - o data protection
 - o copyright
 - o computer misuse
- how moral and ethical issues affect business computer users, i.e.:
 - o the use and abuse of personal and private data
 - o cyberbullying
 - o monitoring of individuals by organisations through the use of:
 - worker monitoring/logging,
 - cookies,
 - key logging,
 - worker call monitoring/recording,
 - electronic consumer surveillance,
 - mobile phone triangulation
- the implications and consequences for organisations of data loss, corruption and theft, i.e.:
 - o legal implications (e.g. action from the Information Commissioner)
 - o impact on customers (e.g. reduced confidence in business, increased risk of personal identity theft)
 - o impact on employees (e.g. disciplinary action for not following company procedures)
 - o impact on organisation (e.g. increased costs in resolving problems caused, loss of income if customers lose confidence)
- the main threats to data security and how to deal with them, i.e.:
 - o threats to data security, i.e.:
 - computer viruses
 - trojans
 - worms
 - phishing
 - spyware
 - adware
 - hacking
 - Denial of Service (DoS) attacks
 - physical threats (e.g. loss/theft of devices)
 - o actions to minimise risks, i.e.:
 - act online in ways which reduce the risk of identity theft and protect personal security
 - use of protection software, i.e. firewall, anti-virus, anti-spam, data encryption to store and transfer data
- using automatic and manual updating facilities for operating systems and security software.

Assessment guidance

During the external assessment, learners will be expected to demonstrate their understanding through questions that require the skills of analysis and evaluation in particular contexts.

2.3 Unit R002: *Using ICT to create solutions in a business environment*

Aims

This unit will enable learners to develop ICT skills that will equip them to operate effectively in a business environment. This unit complements unit R001. In unit R001 learners will study the computer system on which applications software sits and consider the implications of working with data to create content, while in this unit they will work with 'office' applications software to edit and format/create content to meet specified business purposes.

Learners will use a wide range of applications that are commonly used in the workplace, schools, and in further and higher education. They will learn how to select the most appropriate software to complete tasks to meet specified business requirements in a variety of contexts.

They will learn how to use software tools to handle data and communicate information for a range of business purposes, and how to apply formatting to enhance those documents to suit their purpose and intended audience. This type of skill is very valuable as it can be transferred from one software application to another. So, if the learner is able to secure these skills through this unit they will be prepared to use a range of software applications effectively. They will learn to work with a variety of file types and to integrate/import files of different types into other documents. They will develop techniques to search for, select and store information in a variety of contexts. They will learn how to select the tools and techniques to communicate information and solve problems.

On completion of this unit learners will have extended their capability in the use of applications software.

Learning Outcome 1: Be able to use techniques to search for, store and share information

Learners must be taught how to:

- use search engine techniques to find specific information on the internet, i.e. using
 - o phrase
 - o key words
 - o advanced search pages
 - o quotes
 - o wildcards
- use and organise bookmarks/favourites
- select, capture and store graphics and text in compliance with copyright
 - o download
 - o 'copy and paste'
- use non-internet based sources to find information, i.e. local area network, wireless area network, CD-ROMs
- evaluate validity of information, i.e.:
 - o reliability of source
 - o age
 - o bias of information
- reference all information copied/sourced, i.e.:
 - o author/source
 - o year created (if available)
 - o title of webpage/web document
 - o date last updated (if available)

- o URL
- store electronic information¹, i.e.:
 - o meaningful file and folder names
 - o folder structure
 - o backups
 - o password protection
 - o compressing/zippping files
- use email to communicate with others in business contexts, i.e.:
 - o subject line when composing messages
 - o body message text when sending, forwarding and replying to messages, appropriateness of body text
 - o including multiple recipients, i.e.:
 - Carbon Copy (cc)
 - Blind Carbon Copy (bcc)
 - Groups
 - o attachments
 - o email signatures
 - o auto response messages
 - o folders to store messages.

Learning Outcome 2: Be able to select and use software to handle data

Learners must be taught how to use software to handle data, i.e.:

Creating business **spreadsheets**:

- Import/open csv files and save in an appropriate file type
- enter title, column headings and row labels
- enter²/import data, i.e.:
 - o text
 - o numeric
 - o title
 - o column/field headings
 - o row labels
 - o formulae involving arithmetic operators², i.e. +, −, *, /
 - o simple functions², i.e.:
 - SUM
 - AVERAGE
 - MIN
 - MAX
 - IF
 - o cell references, i.e. relative, absolute

Editing and manipulating data in **spreadsheets**:

- insert and delete rows/columns
- change/amend data in cells
- amend formulae
- change data to model outcomes
- search data
- sort data²

- create graphs², i.e.:
 - o pie chart
 - o column/bar graph
 - o line graph
 - o scatter graph

Printing data from **spreadsheets**:

- spreadsheet view
- formulae view
- selected data

Creating flat file (single table) **databases**:

- import csv files and save in an appropriate file type

Editing **databases**:

- enter new records
- edit records
- delete records
- sort table
- query data in a single table using³:
 - o simple criteria, i.e. =
 - o complex criteria, i.e. <, >, <>, >=, <=, NOT, AND, OR, BETWEEN
 - o sort data

Printing **databases**:

- selected data (queries)
- reports
 - o tabular
 - o columnar
 - o stepped
 - o list
 - o label
 - o grouped
 - o grouped with summaries.

Learners must be taught how the purpose and audience for the business activity influences the choice of software.

Teaching should be delivered in the context of data handling software, i.e. spreadsheets and databases.

Learning Outcome 3: Be able to select and use software to communicate information for a business purpose

Learners must be taught how to:

- import txt and rtf files and save in an appropriate file type to retain formatting
- use tools and facilities appropriate to the software, i.e.
 - o enter text, tables, images using the keyboard, mouse or other input device
 - o modify existing documents
 - o design and create new documents
 - o create screen layouts by using existing templates and by creating and positioning text and graphic frames

- o graphics, i.e. copy, paste, resize
- o select appropriate text and graphics
- o edit using insert, delete, cut, copy and paste functions
- o import tables, graphic images, and graphs/charts created in other software
- o integrate files of different types
- o mail merge, i.e.
 - enter merge fields/codes
 - merge selected data
- o use spelling, grammar and design checkers
- proof read documents to detect errors not corrected by automated checkers available within the software used, i.e. spelling, grammar, design checkers.

Learners must be taught how the purpose and audience influences the choice of document type, and how the document type influences the choice of software.

Teaching should be delivered in the context of a range of software, i.e. Word Processing, Desktop Publishing, Presentation, Web page, Graphics, for a range of documents that are typically used in business.

Learning Outcome 4: Be able to use software tools to format information

Learners must be taught how to:

- use headings, subheading and body text
- use widows and orphans
- use white space
- use case, i.e. capitals and lower case and how to use it consistently
- use spacing before and after punctuation, bullets and numbered lists
- use line, paragraph and page breaks and how to use it consistently
- auto date format, i.e. English UK
- use formatting techniques to create impact and enhance the appearance of documents, i.e.:
 - o orientation, i.e. landscape and portrait
 - o margins
 - o inserting page and line breaks
 - o columns
 - o graphics, i.e. positioning, scaling maintaining aspect ratio, cropping
 - o text, i.e. font², style, size, direction, colour², emphasis
 - o paragraph, i.e. alignment, indents, line spacing, tabs
 - o bulleted and numbered lists
 - o text wrapping around objects
 - o cells, i.e.:
 - text, i.e. font, style, size, alignment,
 - number, i.e. decimal places, percentage, currency, date/time
 - o borders and shading²
 - o backgrounds, i.e. images, colours²
 - o transition and animation effects
 - o scaling/fit to page
 - o set print options appropriate to the software²
 - o inserting headers and footers²
- inserting automatic fields, i.e. date and document information.

Connections between units for synoptic assessment

¹ If learners have already completed unit R006 they will have covered the content marked with ¹.

² If learners have already completed unit R003 they will have covered the content marked with ².

³ If learners have already completed unit R004 they will have covered the content marked with ³.

There is no requirement to teach the units in a particular order but teachers should take note of this coverage and schedule the programme of learning accordingly.

2.4 Unit R003: Handling data using spreadsheets

Aims

This unit builds on Units R001 and R002 and learners will be able to apply the skills, knowledge and understanding developed in those units and vice versa. This unit will help the learner to process and present data into meaningful information that can be used to support the decision-making process in real life scenarios.

The learning is important because spreadsheets are used extensively in businesses for a variety of purposes such as budgeting, cost modelling, reporting, trend analysis and forecasting. Spreadsheets are very effective at performing automatic calculations e.g. for displaying information to highlight relationships, for predicting outcomes by changing data, for numerical analyses and to create informative graphs and charts. In a school environment, teachers use spreadsheets to monitor and analyse learners' performance. Learners will create a spreadsheet to provide a solution to a given scenario which will provide the facility for the user to create and manipulate data and to produce graphs and/or charts to support decision making.

On completion of this unit learners will be able to interpret requirements for a spreadsheet, take unstructured data, plan how to use it in a spreadsheet, create and populate a spreadsheet, use relevant functions and tools to manipulate the data and produce outputs to present the data graphically to support decision making.

Learning Outcome 1: Be able to create and populate spreadsheets to meet user requirements

Learners must be taught how to:

- provide structure¹, i.e. worksheet name(s), column/row headings/sub-headings
- set size of rows, columns, print area
- use presentation features, i.e. borders, shading, fonts, colours, wrap text, merge cells, hide/show columns
- use validation rules¹, i.e. list, time, text length
- use input messages and error messages to advise and/or redirect the user in the event of invalid data entry
- enter given data
- set data types¹, i.e. alphanumeric, text, numeric (integer, currency, percentages, number of decimal places and fractions), date/time, limited choice (drop-down list, radio buttons, tick lists), object, logical/Boolean (Yes/No, Male/Female) types
- use conditional formatting
- use page layout features, i.e. headers, footers, gridlines, scaling, paper size, orientation
- provide help for user of spreadsheet (e.g. notes, comments)
- interpret user requirements from the user brief.

Learning Outcome 2: Be able to select and use spreadsheet functions to meet user requirements

Learners must be taught how to:

- select and enter formulae using correct and appropriate arithmetic operators², i.e. +, -, /, *
- select and enter functions², i.e. SUM, AVERAGE, LOOK UPS, IF, AND, OR, TODAY, DATE, MIN, MAX, COUNT
- use absolute and relative cell references

- use brackets to force order of operation
- reference data on other worksheets with formulae, i.e. cell linking
- create, assign, print and annotate macros
- test spreadsheet functionality.

Learning Outcome 3: Be able to use spreadsheet models to present information to support decision making

Learners must be taught how to:

- arrange data by sorting
- reduce data by filtering
- use 'what if analysis' (e.g. trial and error, change variable data, goal seek) to predict results and create different scenarios and outcomes
- recognise that data type influences the graphical method used³
- use graphical methods to present information, i.e. bar chart, pie chart, line graph, scattergram and the use of scales, a title, axis title and key legend.³

Connections between units for synoptic assessment

¹ Unit R001 LO2 supports the development of these skills by developing an understanding of them in business contexts.

² Unit R002 LO2 supports the development of this Learning Outcome.

³ Unit R002 LO2 supports this by developing an understanding of appropriate chart types.

If learners have already completed core units R001 and R002 they will already understand why and how spreadsheets are used to store and retrieve data in a business context and they will have covered the Data Protection Act (DPA). It is therefore recommended that they do these units first. However, if this is not possible teachers should cover the content from the mandatory units which specifically relates to spreadsheets and the DPA as well as the taught content for unit R003 listed above.

2.5 Unit R004: *Handling data using databases*

Aims

This unit builds on Units R001 and R002 and learners will be able to apply the skills, knowledge and understanding developed in those units and vice versa. This unit will enable learners to gain the necessary additional skills and knowledge to be able to modify an existing database by adding fields and then to further enhance a database by creating new table structures to produce a relational database structure. They will also learn how to test and interrogate a database. They will understand that a database has to be developed to meet the needs of an individual user or organisation.

Database software is one of the most important IT applications programs used by organisations in the 21st Century. Databases are used to store and organise data so that it is easy to find the data again when an organisation or individual wants to do something with the data. For example, a website that sells goods is likely to have a database behind it to store details of all the products the business sells and the account details of its customers. A computer game has a database to store all the options selected and to store the results of each game played, so users can compare their progress. A school keeps a database of contact details and exam results of its current pupils.

On completion of this unit learners will be able to modify an existing database and produce a relational database. They will also be able to create queries to interrogate a database and find specific records and produce reports based on the results of these queries and create a user interface for the database.

Learning Outcome 1: Be able to modify databases to meet user requirements

Learners must be taught how to:

- modify existing databases to add field(s), table(s) to meet user needs
- modify single-table databases into multi-table databases and understand the need/rationale for doing this
- create table structures with key fields (primary keys); field names¹
- use data types, i.e. number, text, date/time, currency, auto-incremented, yes/no; field length¹
- use validation rule, i.e. input masks, presence check, range check, list check; customised error messages¹.

Learning Outcome 2: Be able to produce outputs from databases to meet user requirements

Learners must be taught how to:

- create queries and understand the need/rationale for them², i.e.:
 - o simple queries on a single table using single criteria
 - o complex queries on linked tables using single and multiple criteria
 - o reusable queries, i.e. parameter queries using parameter text
- save queries using appropriate query names²
- output results of queries in reports on screen and formatted to print (e.g. address labels, letters, invoices³)
- customise standard reports (e.g. add labels, adjust field widths, re-position fields, add header/footer³)
- customise reports to reflect house styles (e.g. consistent layouts and formatting³).

Learners must be taught how to:

- enable security such as password protection of databases⁴
- create data input forms and add objects, i.e. fields, text labels, pictures, list boxes, combo boxes, option buttons, command buttons⁵
- create menus using forms, use of wizards, i.e. switchboard manager, use of command buttons for navigation, main menu form to load at start up⁵
- customise interfaces to reflect house styles (e.g. consistent layout and formatting⁵).

Learning Outcome 4: Be able to analyse a databases suitability for a business purpose

Learners must be taught how to:

- test validation rules, data entry forms, queries and reports
- solve problems such as responding to error messages, resolving incorrect results from queries
- create a test plan with headings, i.e. test number, test purpose, test data, expected outcome, actual outcome, modification
- use different test methods, i.e. use of test data, i.e. normal, extreme, erroneous test plans, end user testing, i.e. end user/peer testing of user interface to provide feedback.

Connections between units for synoptic assessment

¹ Unit R001 LO2 supports the development of these skills by developing understanding through their use in business contexts.

² Unit R002 LO2 supports this by developing these skills in the context of existing spreadsheets and databases.

³ Unit R002 LO3 and LO4 support this by developing understanding of how to communicate using business documents, of which these are examples.

⁴ Unit R001 LO4 supports this by developing understanding of the need for security measures and the consequences of data loss.

⁵ Unit R001 LO2 supports this by developing understanding of how data can be captured using forms.

If learners have already completed core units R001 and R002 they will already understand why and how databases are used to store and retrieve data in a business context and they will have covered the Data Protection Act (DPA). It is therefore recommended that they do these units first. However, if this is not possible teachers should cover the content from the mandatory units which specifically relate to databases and the DPA as well as the taught content for unit R004 listed above.

2.6 Unit R005: *Creating an interactive product using multimedia components*

Aims

This unit builds on Units R001 and R002 and learners will be able to apply the skills, knowledge and understanding developed in those units and vice versa.

This unit will enable learners to demonstrate their creative flair by combining multimedia components to create a vibrant, energetic or stimulating www, webpage, or interactive product.

Interactive products are used widely in everyday life; from visiting a website, ordering online products, using mobile phone applications, viewing a presentation, e-learning products or playing computer games.

On completion of this unit learners will be able to show how the interactive product meets both the user needs and extends their capability within the use of applications software such as website development.

Learning Outcome 1: Be able to design interactive products

Learners must be taught how to:

- select and use the software features appropriate to the interactive products to aid in the design process, i.e. website, tablet/mobile phone apps, gaming platforms, presentation software
- identify success criteria, i.e. meeting the client brief, component quality, composition and the extent to which the product is interactive
- select and use different forms of navigation site planning techniques, i.e. mood boards, spider diagrams, mind mapping, site plans, house-style, hand-drawn templates
- source and store multimedia components for inclusion in products and how to make ready where applicable, i.e. source components images, video, sound, animation, scripting, sprites. Storage will necessitate the use of different file types¹, i.e. swf, html, sis, app, exe, xaml, xml, ppt
- understand the implications of legislation on their sources, i.e. Copyright Law; Intellectual Property; photo permissions and releases; acknowledgement and referencing of sources²
- select the applications software dependent on purpose and audience, i.e. web authoring software, game making software, 'App' development software or presentation authoring software.

Learning Outcome 2: Be able to create interactive products containing multimedia components

Learners must be taught how to:

- combine components using techniques (e.g. alternative pathway, user interaction and effects)
- use templates, i.e. master slides, environments, cascading style sheets
- create a navigation system, i.e. navigation bar, buttons, hyperlinks, mouse /keyboard controls, menus and drop down lists, graphical user interface
- set up interaction, i.e. roll over, drag and drop, input form, behaviours (e.g. pop-up messages, shake, fades, and sounds) triggers, collision, scripting, hot spots
- use effects, i.e. transitions, html clock, fade in, fade out, custom animation.

Learners must be taught how to:

- test the product during production and where appropriate review tools and techniques used in line with the success criteria
- test the product post completion against the success criteria using a variety of feedback methods including client feedback.

Connections between units for synoptic assessment

¹ Unit R001 LO2 supports this by developing an understanding of appropriate file types.

² Unit R001 LO4 supports this by developing an understanding of the implications of copyright legislation and the consequences of non-compliance with its provisions.

2.7 Unit R006: *Creating digital images*

Aims

This unit builds on Units R001 and R002 and learners will be able to apply the skills, knowledge and understanding developed in those units and vice versa. This unit will enable learners to acquire the underpinning knowledge and skills to enable them to create, edit, enhance and save different types of digital images.

We live, learn, work and play in a very visual world. Whether we like it or not digital images influence our actions and thoughts – persuading us to buy one product instead of another, instructing us to go this way rather than that, explaining a complicated scientific concept and portraying an emotion or expressing a feeling using powerful digital art. With or without words successful digital images will convey their message effectively so that the viewer receives and understands it – and can then act upon it.

The most famous type of digital image is a logo or brand concept. Large companies will spend hundreds of thousands of pounds on their brand image (such as the London 2012 logo; BBC One re-branding) and may re-brand products many times over their life. For example, Pepsi has had 11 re-brandings, showing that the graphic design industry is big business.

On completion of this unit learners will be able to create a digital image that communicates the intended message effectively, meeting the client's needs, and they will have extended their capability within the use of digital editing software packages.

Learning Outcome 1: Be able to specify a digital image solution for a client's needs

Learners must be taught how to:

- analyse a client brief to determine success criteria: suitability; relevance; measurability
- select and use research methods, i.e. image/thought showers/spider diagrams; interviews/focus groups; questionnaires/surveys; competitor/market research/stakeholder perceptions
- select and use creative design plans, i.e. storyboards; roughs/sketches; design concepts/layouts
- select and use component sources, i.e. image capture (e.g. camera, scanner); hand-drawn design; client provided images; stock images; internet; effect of sourced components on final image quality, i.e. file size, resolution, scalability, noise
- identify the implications of legislation on sources, i.e. Copyright Law; Intellectual Property Rights; photo permissions and releases; acknowledgement and referencing of sources¹.

Learning Outcome 2: Be able to create digital images

Learners must be taught how to:

- select and use software for different purposes², i.e. software for vector-based images; software for bitmap/raster-based images
- set image/canvas size and image resolution for different outputs/output sizes for print and for screen³ (as appropriate to the software)
- use standard software tools to create and edit digital images, i.e. cut, copy, crop, paste; select parts of images; move, align and order components; group/ungroup components; rotate and flip; create lines, curves and shapes, i.e. basic and freehand; change stroke and fill, i.e. colour, thickness, style; draw/paint, i.e. pencil, brush, bucket; insert and edit text, i.e. colour, font, size
- select and use specialised software tools to enhance digital images, i.e.:
 - o filters, i.e. sharpen, blur, noise; colour balance, levels and curves; masks and layers

- o retouching tools, i.e. clone, red eye; trace; edit and combine paths
- o opacity/transparency; transform, scale, rotate and distort
- o text effects, i.e. attach to path; guides/guidelines
- combine components to create complex composite digital images, i.e. multiple-step processes; multi-layering; combine output from different software applications
- feedback/evaluate, i.e. recognising merits and faults of technical features; constructive feedback.

Learning Outcome 3: Be able to store, retrieve and present digital images

Learners must be how to:

- use storage systems, i.e. standard naming conventions; version control; archiving
- use file formats for working files⁴, i.e. native file formats (e.g. AI, CDR, PSD, PSP); standard bitmap-based formats (e.g. TIFF, JPEG, GIF, PNG); vector-based formats (e.g. SVG, EPS, WMF, AI, DPP)
- use file formats for final output, i.e. save and/or export; resolution; colour mode; size (e.g. physical and digital); orientation; optimisation⁵/compression; dependent upon method of display or printing
- recognise the effect of different file formats on image quality and size
- use presentation methods⁶, i.e. exhibition; printed portfolio; digital portfolio; mock-up/visual representation; print/web sizes; print media; colour options.

Connections between units for synoptic assessment

If learners have already completed units R001 and R002 they will already understand why and how storage systems are used. Teachers should take note that coverage of storage systems is in units R001, R002 and R006 and manage the teaching of this area of learning accordingly.

¹ Unit R002 LO1 develops research skills in using the internet.

² Unit R001 LO4 supports this by developing an understanding of the implications of legislation including copyright laws and the consequence of non-compliance with their provisions.

³ Unit R002 LO3 supports this by considering how the purpose and audience influences the choice of product and content.

⁴ Unit R001 LO2 supports this by developing an understanding of optimisation and the factors to be taken into account whilst optimising objects.

⁵ Unit R001 LO2 develops an understanding of optimisation and filetypes that addresses these three bullets.

⁶ Unit R002 LO3 supports this by considering how the purpose and audience influences the choice of document type, and how the document type influences the choice of software.

2.8 Unit R007: *Creating dynamic products using sound and vision*

Aims

This unit builds on Units R001 and R002 and learners will be able to apply the skills, knowledge and understanding developed in those units and vice versa.

21st century technology such as gaming technologies, mobile phone apps, media marketing technologies and web-based technologies make great use of dynamic specialist technologies to differentiate their products for the end user. This unit will enable learners to develop the knowledge, understanding and skills that would be expected in creative media industries such as advertising, music and online marketing. Learners will have the opportunity to learn about dynamic products such as music recordings and/or mixes; video/media news clips and animation for webpages before going on to create their own dynamic product.

Movies, animations and sounds bring messages and communications to life. The ability to create, edit and enhance these types of media is an essential business and personal skill that can be used for many purposes. On completion of this unit learners will be able to prepare, create, export and evaluate a timeline-based dynamic product.

Learning Outcome 1: Be able to prepare for the production of dynamic products

Learners must be taught how to:

- analyse a client brief to determine success criteria, i.e. length, file type, main features, theme, message
- produce a time-line storyboard and script for a product suitable to audience and purpose provided by a client
- select appropriate software¹
- source and select appropriate assets and store efficiently
- identify the implications of legislation on sources, i.e. Copyright Law; Intellectual Property Rights; photo permissions and releases (e.g. model, property/location); acknowledgement and referencing of sources.^{2,3}

Learning Outcome 2: Be able to create dynamic products

Learners must be taught how to:

- import sourced assets
- use features of the software to edit the assets, i.e.:
 - o sound, i.e. cut, copy and paste, envelope tool, generate silence
 - o animation, i.e. frame-by-frame, layers, tweening, resizing, rotate, drawing tools
 - o movie, i.e. splitting, layers, mute, sound, titles, audio
- use the features of the software to enhance the assets, i.e.:
 - o sound, i.e. level, fade, amplify, bass, filter, noise removal, reverse, fade, pitch, invert
 - o animation, i.e. motion path, rotation, flip, magnifying, gradient, mask, 3D, transparency, interactions, tints
 - o movie, i.e. transitions, pan, zoom, dissolve, brightness, colour, fade, filter, spin, blur, mirror, edge detection
- combine assets within the selected software to produce an audio, animation or movie suitable to audience and purpose⁴
- save file in original format

- export the product file type suitable for purpose⁵, i.e.:
 - o sound, i.e. WAV, WMA and MP3
 - o animation, i.e. Animated GIF, SWF and Silverlight
 - o movie, i.e. AVI, MP4 and WMV
- recognise the advantages and disadvantages of exporting as different file types, file sizes, compression techniques, optimisation⁶ techniques, codecs, resolution and compatibility.

Learning Outcome 3: Be able to test functionality of dynamic products

Learners must be taught how to:

- create test plans, i.e.:
 - o tests required and how to carry them out
 - o retests
- test the product during production and where appropriate review tools and techniques used in line with the success criteria
- test the product post completion against the success criteria.

Connections between units for synoptic assessment

¹ Unit R002 LO3 supports this by considering how the purpose and audience influences the choice of product and content.

² Unit R002 LO1 supports this by developing an understanding of how to select, capture and store graphics and text in compliance with copyright.

³ Unit R001 LO4 supports this by developing an understanding of the implications of legislation including copyright laws and the consequence of non-compliance with their provisions.

⁴ Unit R002 LO3 develops an understanding of the importance of purpose and audience when editing content.

⁵ Unit R001 LO1 develops an understanding of appropriate filetypes.

⁶ Unit R001 LO2 develops an understanding of optimisation.